Sarcasm Aware Sentiment Analysis

Using Contextual Contrast Detection

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The Problem

- Sarcasm flips the meaning of a sentence. The surface words say one thing, but the writer means the opposite.
- For example, the sentence "I love being stuck in traffic" looks positive but is clearly negative.
- Sentiment models that read only the surface words give the wrong answer on sarcastic input. This affects product reviews, social media monitoring, and customer feedback.

Three Research Questions

- How do different model families compare on sarcasm detection, from classical baselines to fine tuned transformers?
- Are the popular benchmark datasets giving honest performance numbers, or do they have hidden shortcuts that inflate accuracy?
- Can a small set of linguistic contrast features, based on sentiment polarity within a sentence, improve performance and interpretability?

Datasets

- SARC, the Self Annotated Reddit Corpus. About one million Reddit comments, perfectly class balanced, with parent comment context. Labels come from writers themselves.
- Twitter Sarcasm Dataset. Around 43,000 tweets after binarizing to sarcasm versus regular. Smaller and more stylistically uniform than SARC.
- Both datasets are split 80, 10, 10 stratified on the label, with random seed 42 for reproducibility.

Six Model Families Trained

- Classical baselines: Logistic Regression, Linear SVM, and Naive Bayes on TF IDF features.
- Recurrent baseline: a bidirectional LSTM with GloVe initialized embeddings.
- Transformers: DistilBERT and RoBERTa base. RoBERTa is paired with the parent comment on SARC, which is the headline architectural choice of the project.

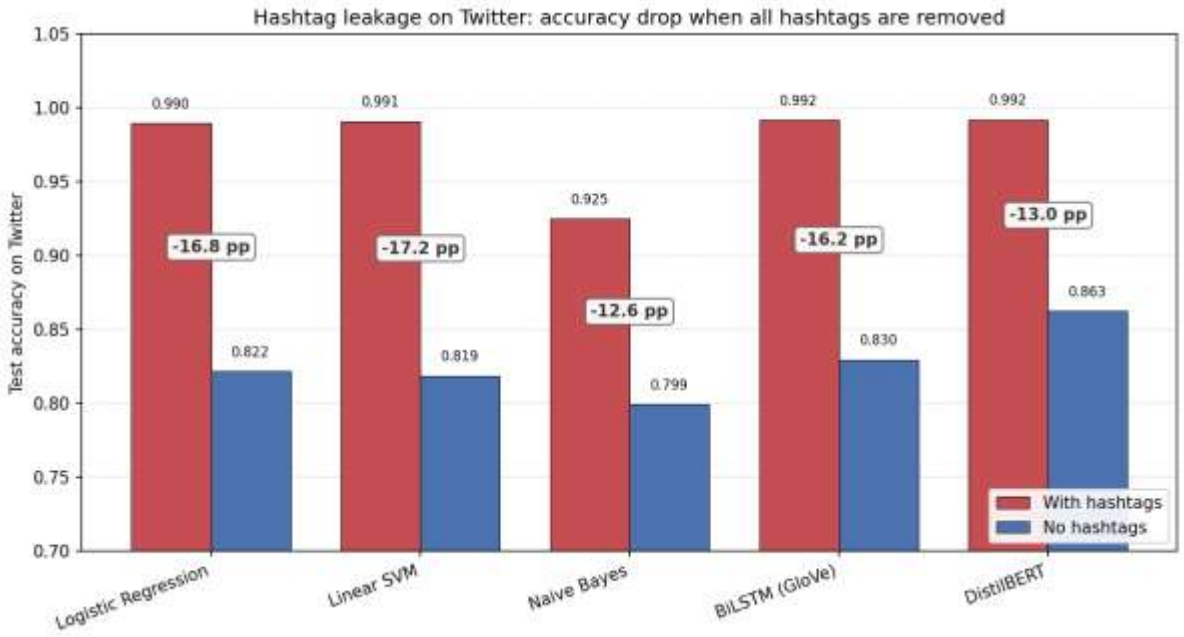
Headline Results

- RoBERTa with parent context is the best model: 79.87 percent on SARC, 87.22 percent on Twitter without hashtags.
- The ranking is monotonic. Each step up in model capacity gives a real improvement.
- Twitter with hashtags shows accuracy above 99 percent for every model, which is too good to be true. The next slide explains why.

Model	SARC	Twitter (with #)	Twitter (no #)
Logistic Regression	0.7238	0.9898	0.8218
Linear SVM	0.7230	0.9910	0.8187
Naive Bayes	0.6873	0.9250	0.7993
BiLSTM with GloVe	0.7514	0.9917	0.8296
DistilBERT	0.7760	0.9921	0.8625
RoBERTa (parent context)	0.7987	n/a	0.8722

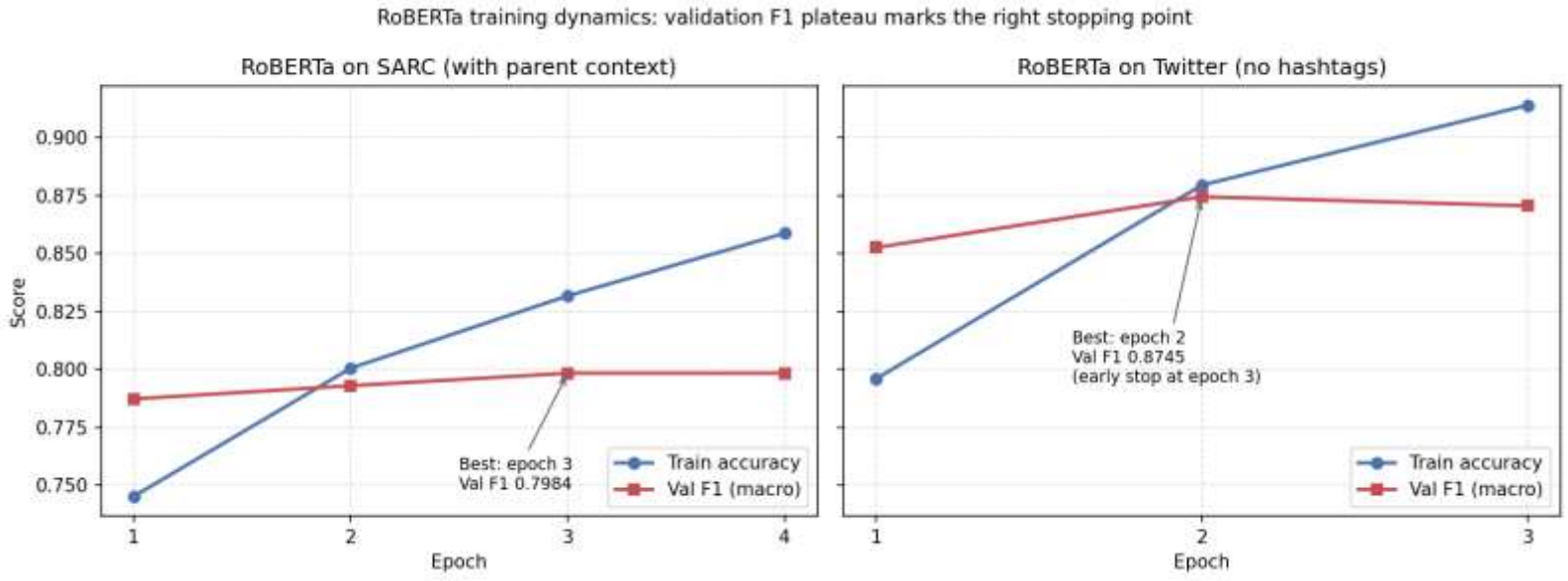
The Hashtag Leakage Finding

- Stripping all hashtags from Twitter drops accuracy by 13 to 17 percentage points across every model family.
- The drop is consistent across classical, recurrent, and transformer architectures, which means leakage is a property of the data, not any single model.
- The widely reported 99 percent accuracy on Twitter sarcasm is largely a dataset artifact. Honest evaluation gives 80 to 87 percent.



RoBERTa with Parent Context

- On SARC, the parent comment is paired with the target comment using the standard separator format. This gives RoBERTa the context a human reader would have.
- Result: 79.87 percent test accuracy on SARC, a gain of 2.27 points over DistilBERT. Conversational context is the strongest single signal for Reddit sarcasm.
- Training history shows a healthy plateau on SARC and early stopping on Twitter at epoch 3, where the smaller dataset overfits faster.

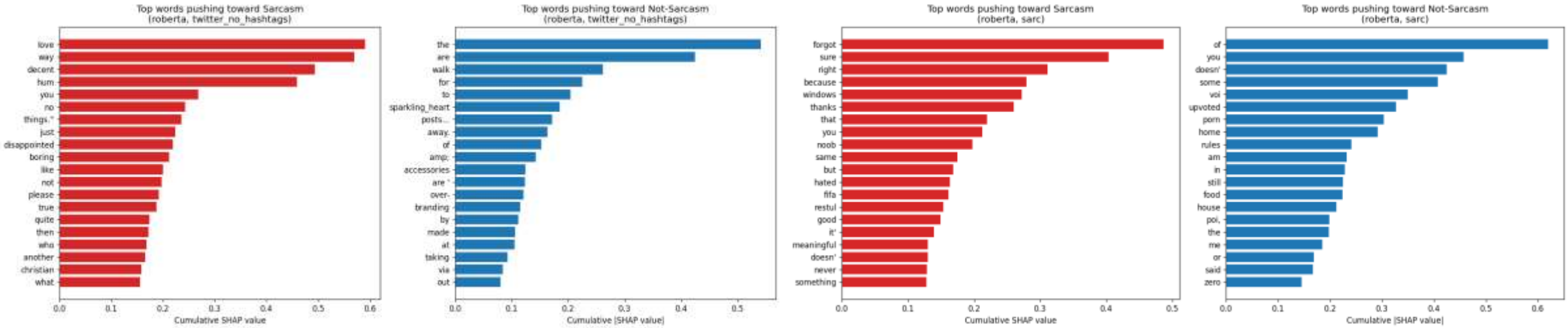


Contrast Features Ablation

- Eleven features were built using VADER sentiment scores: positive score, negative score, polarity swing, word clash, and so on.
- The hypothesis was that sarcasm often mixes positive and negative polarity in the same sentence, like a positive word sitting next to a bad situation.
- Result: deltas range from minus 0.37 to plus 0.16 percentage points. The features do not measurably help on top of strong text representations. This is an honest negative finding.

Interpretability: LIME and SHAP

- LIME and SHAP were applied to RoBERTa on both datasets. Twenty examples explained per dataset, balanced across correct and incorrect predictions.
- On Twitter, both methods agree on 8 out of 15 top sarcasm cues including love, way, decent, boring, like, no, quite, and true. The model has learned a clean ironic intensifier template.
- On SARC, agreement drops to 5 out of 15. The lexical signal is more diffuse because RoBERTa relies more on parent context than on individual words.



Twitter (53 percent agreement)

SARC (33 percent agreement)

Failure Modes

- The phrase "i love you" is predicted as sarcasm with 81.5 percent confidence by the Twitter model. Negation does not help: "i do not love being woken up at 5am" is also predicted as sarcasm with 98.5 percent confidence.
- Reason: the word love is the single highest weighted sarcasm cue. In Twitter training data, almost every occurrence of love is in an ironic context. The model has learned a shortcut, not the concept of sarcasm.
- The SARC model behaves correctly on the same input, which proves the failure is corpus specific, not architecture specific.

Future Work

- Hard negative augmentation. Add around 5,000 sincere uses of love and other intensifiers to the Twitter training set to break the lexical shortcut.
- Explicit negation handling. Append a NEG marker to tokens following negation words, so the model treats do not love and love as different inputs.
- Larger and more diverse training corpus, combining SARC, Twitter, and conversational datasets like MUStARD. Apply the parent context idea to Twitter using thread reconstruction.

Summary

- Best result: RoBERTa with parent context reaches 79.87 percent on SARC, the strongest score among the architectures tried.
- Most important methodological finding: hashtag leakage on Twitter inflates accuracy by 13 to 17 percentage points. Honest evaluation requires the no hashtags ablation.
- Most important interpretability finding: what the model calls sarcasm is whatever its training corpus calls sarcasm. There is no general sarcasm classifier here, only two corpus specific ones.

Thank you. Questions?